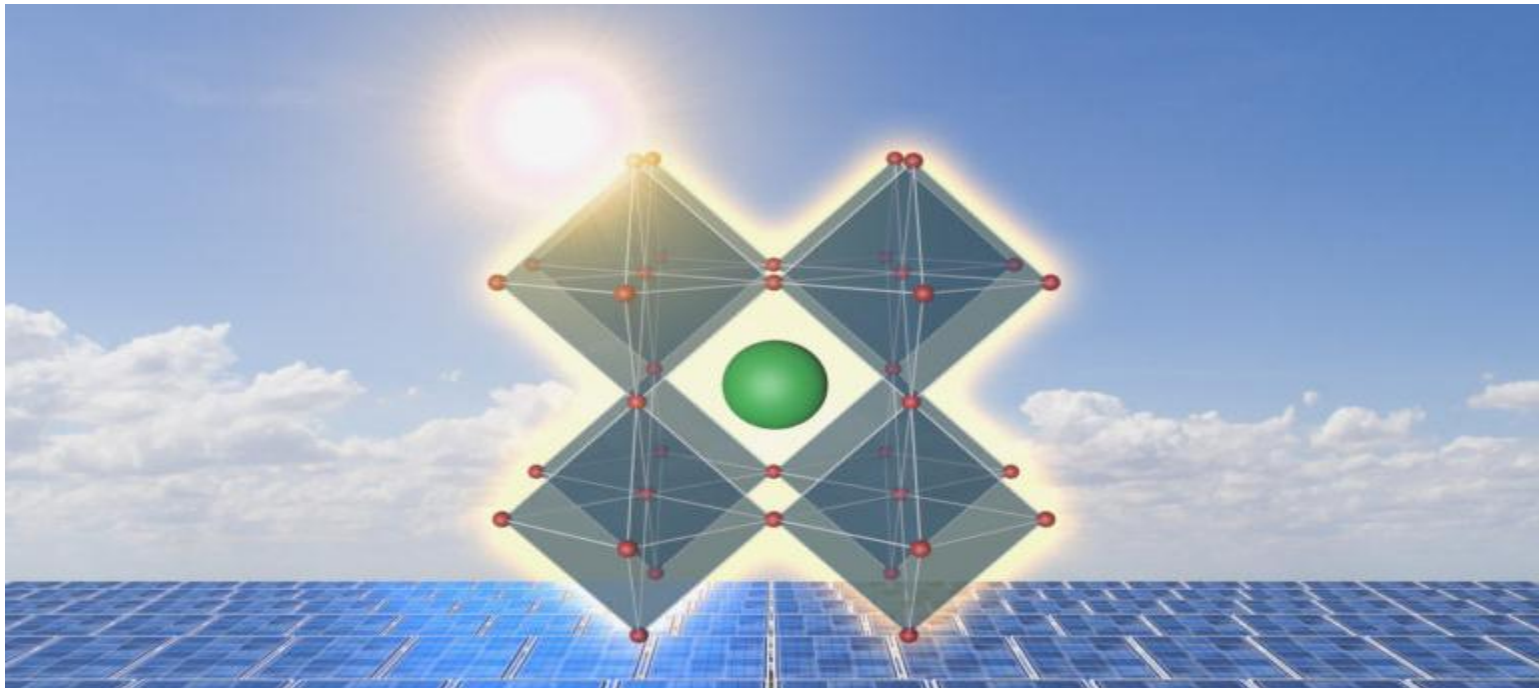


LAB: 2

Fabrication and Characterization of Perovskite Solar Cells

Instructor: A/P Aung Ko Ko Kyaw

Lab Demonstrator: Dr. Aamir Saeed





Content

- Composition of Perovskite: Normal bandgap (1.56 eV)/ Wide bandgap (1.84 eV)
- Device Structure
- Preparation of Solutions
- Fabrication Steps
- Power Conversion Efficiency (PCE) Calculation



Compositions of Perovskite

Normal Bandgap (1.56 eV)



Precursors

1. Cesium iodide (CsI)
2. Formamidinium iodide (FAI)
3. Methylammonium bromine (MABr)
4. Lead iodide (PbI_2)
5. Lead bromide (PbBr_2)
6. N-dimethylformamide (DMF) (Solvent)
7. Dimethyl sulfoxide (DMSO) (Solvent)

Wide Bandgap (1.84 eV)

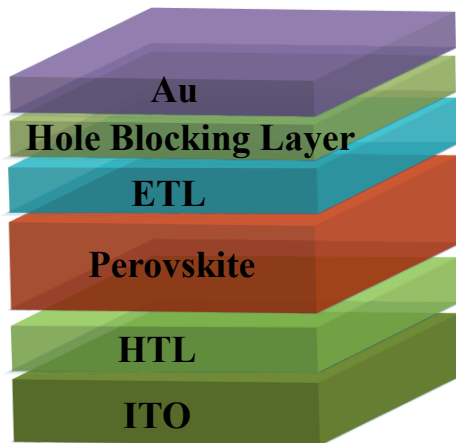


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Device Structure

Inverted positive-intrinsic-negative (**p-i-n**)



- ❖ ITO/glass substrates
- ❖ Hole Transport Layer (HTL)
- ❖ Perovskite Absorber Layer
- ❖ Electron Transport Layer (ETL)
- ❖ Hole Blocking Layer
- ❖ Au Electrodes



Solution Preparation

Perovskite Precursor Preparation

181.4 mg FAI, 6.3 mg MABr, 16.8 mg CsI, 2.03 mg PbBr_2 , and 529.0 mg PbI_2 were dissolved in 1 mL mixed solvents of **DMF: DMSO = 4:1 (vol/vol)**.

HTL

Self-assembled monolayer (SAM) 4PADCB solution (0.5 mg mL^{-1} in **ethanol**).

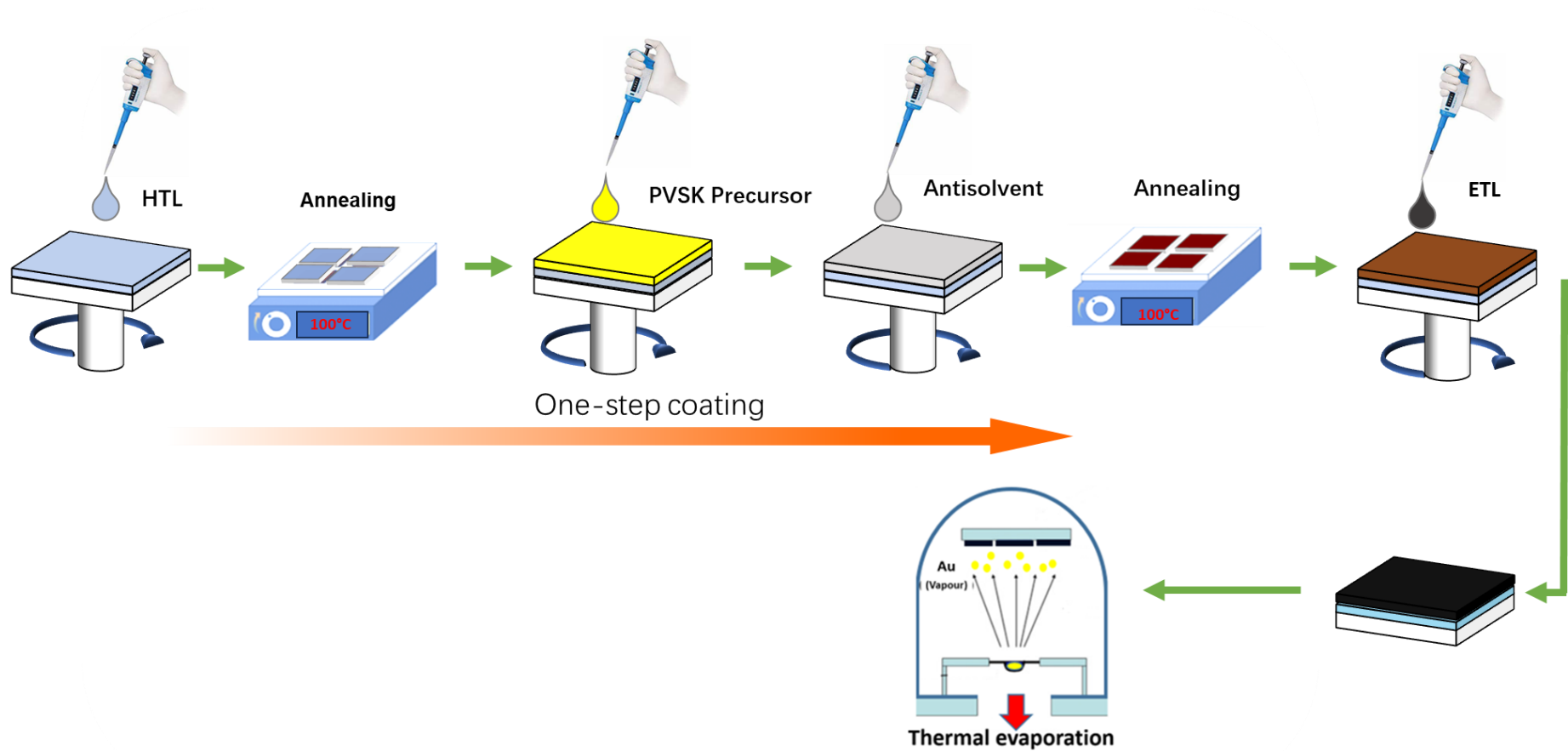
ETL

(6,6)-phenyl-C61-butyric acid methyl ester (PC61BM) was obtained by spin-coating the PC61BM solution (20 mg mL^{-1} in **Chlorobenzene, CB**).

Hole Blocking Layer

2,9-dimethyl-4,7-diphenyl-1,10-phenanthroline (BCP) (0.5 mg/ml in **isopropanol**).

Fabrication Steps





Device Fabrication Steps

- 1) ITO/glass substrates ($1.5\text{ cm} \times 1.5\text{ cm}$, $9\ \Omega/\text{sq}$, Advanced Election Technology) are sequentially ultrasonically cleaned in diluted detergent, deionized water (DI), isopropanol (IPA), and ethyl alcohol for 15 min.
- 2) The pre-clean ITO glasses are treated with UV-Ozone for 25 min before depositing the HTL.
- 3) **HTL:** The SAM (4PADCB) is spin-coated statically at a speed of 4000 rpm (for 30 s) annealed at $120\text{ }^{\circ}\text{C}$ for 10 min.
- 4) **Perovskite Deposition:** $50\ \mu\text{L}$ of the precursor solution is spread onto the HTL substrate, and spin-coated using a two-stage procedure (1000 rpm for 5 s and 4000 rpm for 30 s), with $160\ \mu\text{L}$ of CB as antisolvent dripped at 11s of the second stage. The as-prepared perovskite films are annealed at $100\text{ }^{\circ}\text{C}$ for 30 min.
- 5) **ETL:** The ETL is obtained by spin-coating the PC61BM solution (20 mg mL^{-1} in CB) on the substrate (3000 rpm , 30 s).
- 6) **Hole Blocking Layer:** Spin-coated BCP (0.5 mg mL^{-1} in IPA) at 5000 rpm for 30s.
- 7) **Electrode Deposition:** The whole device is transferred to a vacuum chamber under a base pressure of $5 \times 10^{-4}\text{ torr}$. 100 nm Ag is sequentially thermal evaporated with a shadow mask area of 0.04 cm^2

Power Conversion Efficiency (PCE) Calculation

The J-V curves of the perovskite solar cells are measured with a Keithley 2400 source meter under 1 sun (AM 1.5G spectrum) generated from a class solar simulator (Japan, SAN-EI, XES-40S1).

Important Parameters:

- ❖ Short-circuit current density (J_{SC})
- ❖ Open-circuit voltage (V_{OC})
- ❖ Fill factor (FF)
- ❖ Power Conversion Efficiency (PCE)

EQE Measurement: EQE spectrum is measured using a Solar Cell Spectral Response Measurement System QE-R3011 (Enlitech Co., Ltd.). The light intensity at each wavelength is calibrated using a standard single-crystal Si photovoltaic cell.



Discussion topics should be included in the lab report

- What are the functions of each layer in the perovskite solar cell?
- What is the purpose of anti-solvent during the perovskite film deposition?
- Why does the normal bandgap perovskite give higher PCE and J_{SC} , but lower V_{OC} than the wide bandgap perovskite?
- What is the application of wide-bandgap perovskite solar cell?
- What are the estimated bandgaps of the perovskites based on the measured EQE spectra?